

QuranicMMLU: A Cognitively-Aware Benchmark for Evaluating Generative AI Solutions on Quranic Linguistic Knowledge

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Abstract

We introduce QuranicMMLU, a benchmark for evaluating generative AI on Quranic Arabic across multiple dimensions of linguistic complexity. Existing Quranic benchmarks center on general question answering and semantic retrieval, without probing specific linguistic competencies or stratifying by cognitive demand and verse difficulty. We construct a five-pillar Quranic taxonomy spanning Phonology, Morphology, Syntax, Semantics, and Pragmatics, with 31 leaves covering phenomena from tajwīd and root-and-pattern morphology to occasions of revelation and inter-surah coherence. For each leaf we generate questions stratified by Bloom’s cognitive level and verse perplexity, then have LLM as a judge to independently answer and score every item and route the annotations to manual review. The resulting dataset comprises 980 human-reviewed questions, each issued in both open-ended and multiple-choice form. We benchmark 12 systems on these items and find that the Islamic-specialized model leads, yet every system scores higher on multiple-choice accuracy (average 84%) than open-ended answer quality (average 60%): the two rankings agree closely (Kendall’s $\tau = 0.73$), but multiple-choice scoring hides failures that surface only once answer choices are removed. QuranicMMLU thus offers a rigorous, linguistically grounded framework for evaluating Arabic NLP in the Quranic domain.

1 Introduction

The Quran is the primary religious and cultural text for nearly two billion Muslims, and engaging with it meaningfully demands not just reading but a genuine understanding of its language and content. As large language models are increasingly consulted on Islamic topics, including direct questions about the Quran (Abdelaal et al., 2026), the

cost of ungrounded answers in this domain is unusually high: hallucinated or inaccurate responses carry real religious consequences (Mubarak et al., 2025). Reliable use of these systems therefore depends on accurate, fine-grained evaluation of how well they actually understand Quranic Arabic.

Existing benchmarks only partially meet this need. Islamic and Quranic QA resources assess factual and reasoning ability (Abdelaal et al., 2026). Three gaps remain. First, no benchmark targets the diversity of *linguistic* competencies, from phonology to pragmatics, that Quranic understanding requires. Second, most datasets are multiple-choice only, a format that can be gamed through answer-choice artifacts (Balepur et al., 2024) and that diverges from how users actually query models. Third, the cognitive demand of questions is rarely measured systematically.

We introduce QuranicMMLU to address these gaps. Our contributions are:

- (i) A five-pillar taxonomy of Quranic Arabic spanning Phonology, Morphology, Syntax, Semantics, and Pragmatics, with 31 leaves covering competencies from tajwīd to inter-surah coherence;
- (ii) A human-reviewed, retrieval-grounded dataset of 980 questions, each in both open-ended and multiple-choice form and stratified by Bloom’s cognitive level (Anderson and Krathwohl, 2001) and verse ‘popularity of use’, measured by perplexity;
- (iii) An evaluation of 12 systems in both open-ended and multiple-choice formats, showing that multiple-choice scoring overstates ability and hides failures that surface only in open-ended use.

The leaderboard and relevant resources are available at <https://huggingface.co/spaces/musiml-org/QuranicMMLU>.

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2 Related Work

2.1 Islamic and Quranic QA Benchmarks

Early Quranic QA frames the task as extractive answer-span retrieval (Malhas and Elsayed, 2022; Malhas et al., 2022, 2023; Alnefaie et al., 2023), while recent benchmarks broaden to Islamic knowledge and reasoning through multiple-choice, retrieval-augmented, and shared-task settings (Abdelaal et al., 2026; Bhatia et al., 2026; Boucekif et al., 2025; Alwajih et al., 2025), and extend to Islamic jurisprudence across the Sunni legal schools (Atif et al., 2025; Bahaj and Ghogho, 2025). These resources target factual and reasoning ability; we instead probe Quranic *linguistic* competencies, from phonology to pragmatics, in both multiple-choice and open-ended formats.

2.2 Arabic Understanding Benchmarks

General Arabic benchmarks such as ArabicMMLU and AlGhafa measure broad competence (Koto et al., 2024; Almazrouei et al., 2023), but not the phonological, morphological, syntactic, and rhetorical phenomena specific to Quranic Arabic that we target.

2.3 Cognitive and Format-Aware Evaluation

Bloom’s taxonomy (Krathwohl, 2002) has become a foundational scaffold for stratifying cognitive demands in language model evaluation, moving assessment beyond aggregate accuracy toward diagnosing where competence breaks down across cognition layers. Recent work has demonstrated the applicability of cognitive assessment frameworks across visual understanding (Abootorabi et al., 2026) and relational knowledge in multilingual contexts (Abbasi Liasi et al., 2026). However, most benchmarks remain limited: they concentrate on lower-order cognitive tasks, rely predominantly on multiple-choice formats vulnerable to answer-choice artifacts (Balepur et al., 2024), and lack systematic frameworks for comprehensive cognitive evaluation. To address these gaps, we introduce QuranicMMLU, a benchmark that combines multiple question formats with a multi-judge protocol to systematically evaluate knowledge across cognitive levels and verse-perplexity dimensions.

3 Approach

QuranicMMLU is built through a six-stage pipeline (Figure 1): we define the taxonomy, generate questions per leaf, collect LLM annotations, conduct

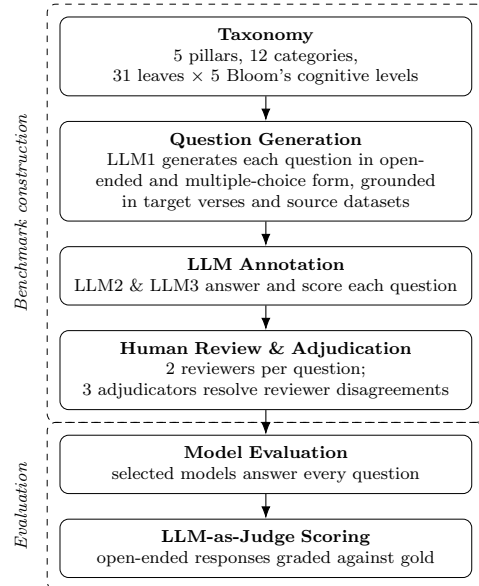


Figure 1: Overview of QuranicMMLU: taxonomy-driven question generation, LLM annotation and human review, then model evaluation with LLM-as-judge scoring. LLM1 is Claude (generator); LLM2 and LLM3 are GPT and Gemini (annotators).

human review with conflict resolution, benchmark selected models, and score open-ended responses with an LLM judge. Each stage is detailed below.

3.1 Taxonomies

QuranicMMLU Taxonomy. We built the taxonomy by studying the literature and key sources for each pillar. We drafted candidate categories and refined them with the help of AI deep-research tools, ensuring broad coverage of the target capability space by including both commonly studied phenomena and less frequently represented but conceptually important aspects. Related items were then grouped into the final taxonomy.

We organize the QuranicMMLU Taxonomy as 5 pillars, 12 categories, and 31 leaves (Figure 5 in Appendix A, p. 11), each leaf a distinct phenomenon, so evaluation is a grid of competencies rather than one aggregate score. Unlike MMLU (Hendrycks et al., 2021) and ArabicMMLU (Koto et al., 2024), which probe subject knowledge, we organize by linguistic ability (Ribeiro et al., 2020), spreading each pillar’s leaves across a range of its sub-phenomena.

Phonology covers madd, qalqalah, waqf, and qirā’āt, probing the sound–meaning mapping that Arabic’s undiacritized script makes a hard disambiguation task (Zitouni et al., 2006). *Morphology* spans roots, lemmas, wazn, tense–aspect–mood, voice, number, and clitics, all sources of the

high ambiguity of the root-and-pattern system and grounded in the Quranic Arabic Corpus (Dukes and Habash, 2010). *Syntax* ranges from sentence type and idāfa to negation scope, fronting, ellipsis, and disputed i'rāb, reflecting the difficulty of Arabic parsing (Green and Manning, 2010) and grounded in the dependency treebank (Dukes et al., 2010). *Semantics*, the widest pillar, covers lexical ambiguity (furūq, wujūh wa naẓā'ir), figurative language, and surah-level meaning, all open problems in Arabic word-sense disambiguation (Jarrar et al., 2023; Raganato et al., 2017), figurative-language understanding (Tsvelkov et al., 2014), and discourse coherence (Barzilay and Lapata, 2008). *Pragmatics* targets speech acts, asbāb al-nuzūl, and cross-verse reasoning, tasks needing communicative-function recognition (Stolcke et al., 2000) and context beyond span retrieval (Malhas and Elsayed, 2022; Malhas et al., 2022). Appendix G (p. 9) gives one representative question per pillar with its verified Bloom level.

Bloom’s Cognitive Level Taxonomy. We stratify questions across five of the six levels of Bloom’s taxonomy, Remember, Understand, Apply, Analyze, and Evaluate, omitting Create because generating new Quranic text raises clear ethical concerns (Figure 6 in Appendix A, p. 12). The operational test reviewers use to assign these levels is given in Appendix F (p. 9).

3.2 Datasets

We ground questions and answers in authoritative sources: the Quran.com API for **Phonology**; the Quranic Arabic Corpus for **Morphology** (Dukes and Habash, 2010) and **Syntax** (Dukes et al., 2010); Quranic tafsīr from Tarteel¹ for **Semantics**; and an asbāb al-nuzūl dataset² based on *Sahih Asbab al-Nuzul* (Ibrahim Muhammad al-Ali) for **Pragmatics**.

3.3 Question Generation and Annotation

Generation. For each leaf, we use Claude Opus 4.7 to generate questions, reference short answers, and multiple-choice options at each Bloom level. Generation begins by selecting a verse or group of verses containing an instance of the target leaf (for example, a verse with a simile or metaphor when asking about figurative language). We build this

pool by searching the grounding datasets for verses that contain an instance of the target phenomenon, at times using AI research tools to help find candidates, and manually reviewing them to confirm the instance is present; generation then draws a verse at random. We also balance questions across verse perplexity (computed using the Gemma-3-12B model), a proxy for the model’s familiarity with a verse, aiming for comparable counts per leaf across the low, medium, and high levels; this stratification is applied to single-verse questions rather than whole-surah ones. The relevant verse is included in the prompt (except for surah-level questions) to avoid misretrieval of Quranic text. Each question is produced with a four-layer prompt stack (Appendix B, p. 8; Figure 7) in which every leaf and Bloom level has a specialized prompt, and each item is written to support both open-ended and multiple-choice answering. Of the four options per question, one is correct, two are close distractors, and one is clearly wrong. For both formats, the datasets (Section 3.2) ground every answer in an authoritative source (a retrieval-augmented setup).

LLM annotation. The generated items are then annotated by two LLM judges, GPT-5.2 and Gemini-2.5-Flash (Comanici et al., 2025). Each judge writes an open-ended answer, selects the correct MCQ option, assigns a Bloom level, and scores the question out of five on specificity, answerability, leaf relevance, and clarity (Appendix C, p. 9). Both judges are given full access to the grounding datasets (Section 3.2) and prompted to base answers on them and cite their source otherwise, again making annotation retrieval-augmented rather than memory-dependent. On the multiple-choice answer, the three independently prompted models (the generator and the two judges) show almost-perfect agreement (Fleiss’s $\kappa = 0.90$).

Human review and adjudication. Every annotation is then reviewed by two humans per question to reduce individual bias. Reviewers are undergraduate students, fluent Arabic speakers with a background in Islamic knowledge; because they are not religious scholars, every decision is grounded in corpus evidence and cited references rather than personal judgment, following a written guide (Appendix F, p. 9) that ranks sources of truth: corpus evidence first, then established scholarly consensus, then genuine dispute. Reviewers may edit the question, correct its Bloom level, choose among the annotated answers, add a new answer when none is correct, or delete items that are irrelevant,

¹<https://huggingface.co/datasets/tarteel-ai/quran-tafsir>

²<https://github.com/mostafaahmed97/asbab-al-nuzul-dataset>

incorrect, or controversial; because we prioritized correctness and Islamic soundness, any item with a suspected hallucination, an unresolved scholarly dispute, or a single flagged concern was discarded when it could not be cleanly fixed. Counting any such edit as a revision, 370 of the 1,086 candidates (34%) were revised, most often a corrected Bloom level (235 items), 106 (10%) were discarded, and 610 were accepted as generated, leaving the final 980. Conflicts were settled by majority verdict among three adjudicators, with 133 items escalated to group discussion. The review interface is shown in Appendix D (p. 9).

Dataset statistics. Figure 4 (Appendix A, p. 10) reports the distribution of the 980 questions, showing the Bloom-level and verse-perplexity composition within each pillar. The dataset spans all five pillars and all five Bloom levels, with Syntax, Semantics, and Morphology the most represented pillars and a deliberate concentration of higher-perplexity verses, so that evaluation is not dominated by verses models have likely memorized.

3.4 Benchmarking

We benchmark the final dataset in both formats, MCQ and open-ended, over the identical item pool, so results are directly comparable across the two. We evaluate two classes of systems: (i) **open-source models**, general-purpose open-weight LLMs including Gemma (Google) (Gemma Team et al., 2025, 2026), Qwen/Qwen-VL (Alibaba) (Yang et al., 2025; Bai et al., 2025), and the Arabic-centric ALLaM (Bari et al., 2024); and (ii) **Islamic models** specialized for Quranic content, namely Fanar-Sadiq, the Islamic-oriented model from Fanar (QCRI) (Fanar Team et al., 2025, 2026), and Ansari (Kadous et al., 2026).

3.5 Metrics

We evaluate the two formats with complementary metrics, all broken down by model and by Bloom level, with full breakdowns in Figure 2. For **open-ended** answers, we use Gemini as an automated grader that scores each response from 0 to 5 given the question, a reference answer, and the model’s response (Appendix E, p. 9); we report this rating as a percentage of the 5-point maximum, together with the proportion of responses scoring 80% or higher. For **MCQ**, we report accuracy, the share of questions answered correctly, presented by Bloom level, by taxonomy content type, and by dataset; with four options, random guessing yields roughly 25%.

Model	MCQ Acc. (%)	Open (%) mean $\pm$ SD
Ansari	96.4	93.2 $\pm$ 21.8
gemma-4-31B	93.3	82.5 $\pm$ 28.8
Qwen3.6-27B	91.8	62.1 $\pm$ 37.0
gemma-4-26B-A4B	91.2	75.7 $\pm$ 32.8
Qwen3.5-35B-A3B	89.7	62.1 $\pm$ 35.8
Qwen3-235B-A22B	88.3	52.3 $\pm$ 39.0
Qwen3.5-9B	84.5	53.1 $\pm$ 36.6
Fanar-Sadiq	80.1	66.7 $\pm$ 37.0
Qwen3-VL-8B	76.3	41.1 $\pm$ 37.2
gemma-3-12B	73.2	50.2 $\pm$ 38.2
Qwen3-VL-4B	69.7	36.0 $\pm$ 36.4
ALLaM-7B	69.5	46.4 $\pm$ 39.8

Table 1: Overall benchmarking results: multiple-choice accuracy and open-ended Gemini score, both as percentages, reported as mean $\pm$ standard deviation across items and sorted by MCQ accuracy. Per-pillar and per-Bloom breakdowns are in Figure 2.

Finally, to test whether the two formats measure the same underlying ability, we compare model rankings under each using Kendall’s τ , where a strong correlation would indicate that both formats probe similar skills.

4 Results

On the **open-ended** evaluation, Gemini judge scores spanned 93.2% for Ansari down to 36.0% for Qwen3-VL-4B (Table 1). Ansari also scored 80% or higher on 91.1% of items, ahead of the next system, gemma-4-31B, which averaged 82.5% and cleared that threshold on 75.6% of items. Among the remaining systems, Fanar-Sadiq placed fourth (66.7%), and the largest model evaluated, Qwen3-235B-A22B, sat in the lower half (52.3%). The same items in **multiple-choice** form produced higher scores for every system: Ansari again placed first (96.4%), followed by gemma-4-31B (93.3%), while ALLaM-7B was the weakest at 69.5%, still well above the 25% chance level, and the seven highest-scoring systems each cleared 84%.

The two orderings were closely aligned (Kendall’s $\tau = +0.7273$; 57 of 66 model pairs concordant). Ansari and gemma-4-31B occupied the top two positions in both, while Fanar-Sadiq moved up four places under the open-ended setting (eighth in MCQ, fourth in open-ended). The full per-pillar and per-Bloom breakdown is shown in Figure 2. Across pillars, Phonology yielded the weakest scores for almost every system in both settings, dropping below 34% for most systems on the open-ended questions, whereas Morphology and Semantics were consistently the strongest. Across

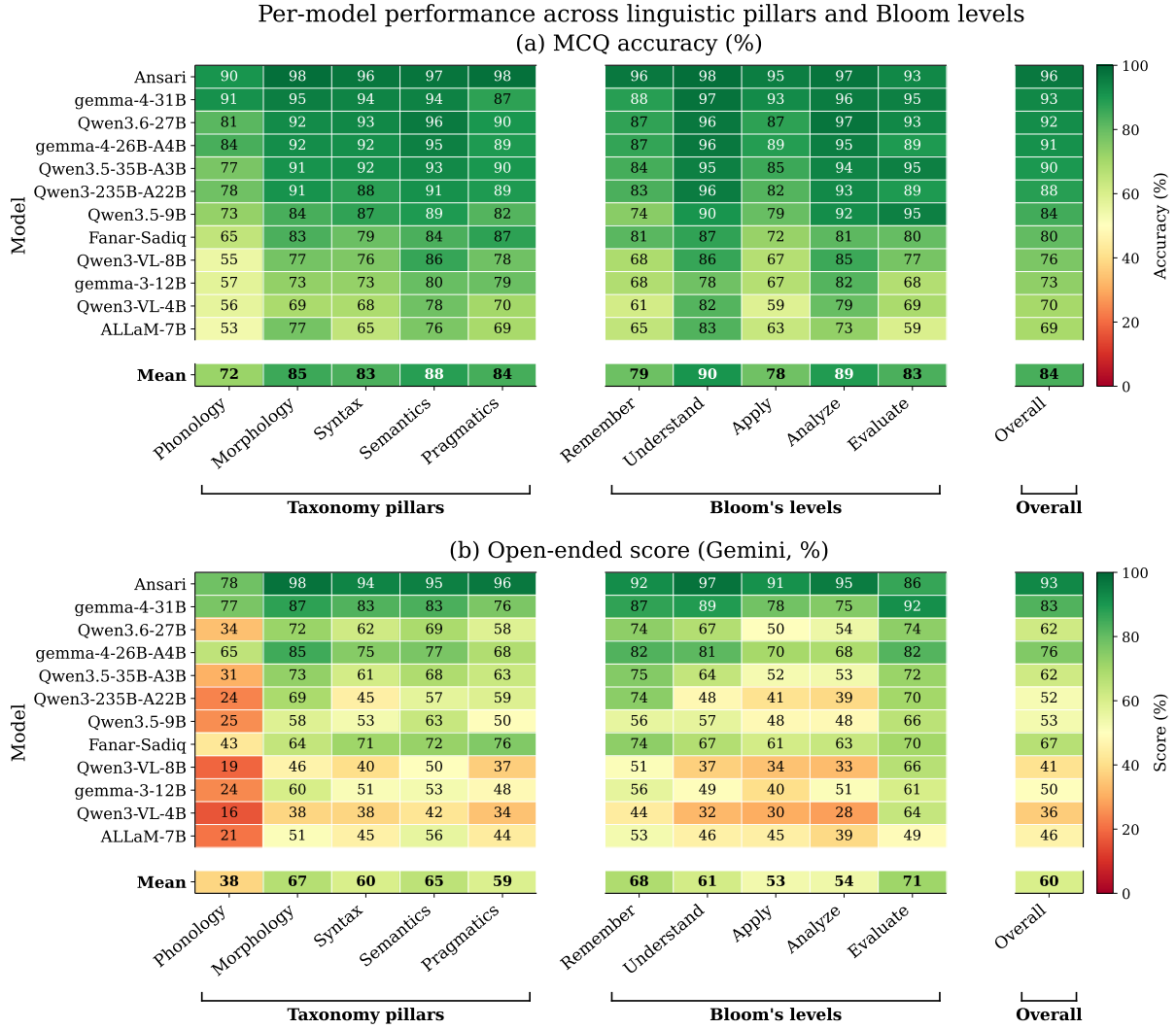


Figure 2: Per-model performance across the five linguistic pillars and the five Bloom levels, for (a) MCQ accuracy and (b) open-ended Gemini score. Models are ordered by overall MCQ accuracy. Phonology is the weakest pillar for nearly every system, especially in the open-ended setting.

Bloom levels, scores peaked at the Understand level and fell to their lowest at Apply and Analyze in both settings.

5 Discussion and Conclusion

Ansari, a system specialized for Islamic and Quranic content, ranked first in both formats, and this domain specialization is reflected directly in our results. The strong rank correlation between formats (Kendall’s $\tau = +0.7273$) indicates that, at the level of ordering systems, multiple-choice accuracy and open-ended quality largely track the same underlying ability, which might suggest that MCQ benchmarks could serve as an efficient proxy, though confirming this would require a dedicated error analysis of the LLM-as-judge scores. The agreement is not perfect, however: Fanar-Sadiq rose four places under open-ended grading not be-

cause it improved in absolute terms but because other systems degraded more sharply when the answer choices were removed, a gap that pure MCQ evaluation would hide.

The Bloom-level pattern mirrors a familiar profile of model cognition: performance was highest at the Understand level and lowest at Apply and Analyze, suggesting that current systems recognize and explain Quranic phenomena more readily than they apply rules to new cases. Overall, QuranicMMLU provides a fine-grained, linguistically grounded view of model ability on Quranic Arabic and shows that format and cognitive demand both shape measured performance. Future work includes broadening leaf coverage, adding human scoring of open-ended answers, and evaluating further Islamic-specialized systems.

6 Limitations

Our study has several limitations. (i) The taxonomy is designed to be broad and to cover a variety of phenomena within each pillar, but it does not capture every linguistic rule or concept in those pillars. (ii) Open-ended responses are scored solely by an LLM judge without human scoring, which may introduce model bias, including a possible preference for answers resembling a judge’s own model family. (iii) We evaluate only open-weight and open-access systems; closed-source commercial models were not assessed because we did not have access to their APIs. (iv) Our benchmark does not yet include every relevant Islamic-specialized system, and some top-ranked models on Islamic-domain leaderboards remain to be evaluated. (v) Questions are generated by a single model (Claude Opus 4.7), so despite human review some generation-model artifacts may remain.

7 Ethical Statement

We note the following ethical considerations. (i) Our human reviewers were not religious scholars; they reviewed questions using the grounding datasets, reference books, online research, and their own judgment. This dataset is intended solely for benchmarking model capabilities and should not be treated as an authoritative source of religious fact. (ii) We omit the Create level of Bloom’s taxonomy because generating new Quranic text raises clear ethical concerns.

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Appendices

The appendices follow the order of the pipeline in Section 3.1ff: the taxonomy (A, p. 8), question generation (B, p. 8), the quality rubric (C, p. 9) and review interface (D, p. 9) used for annotation, the open-ended judge prompt (E, p. 9), the Bloom-level verifier guidelines (F, p. 9), and representative questions per pillar (G, p. 9).

A QuranicMMLU Taxonomy

Figure 5 shows the full taxonomy: 5 pillars, 12 categories, and 31 leaves. Each pillar maps to a classical discipline of Quranic study, each category groups related phenomena, and each leaf defines a single competency that questions target. Figure 6 shows the five Bloom’s cognitive levels along which questions are stratified, and Figure 3 the overall Bloom-level composition of the dataset. Figure 4 breaks that composition down by pillar, together with the verse-perplexity classes.

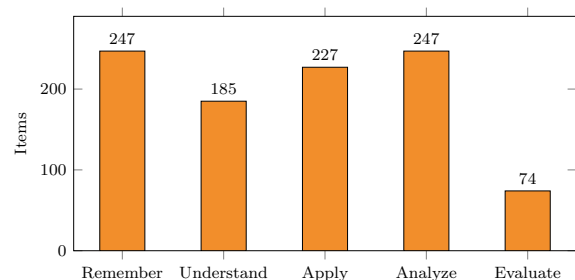


Figure 3: Overall Bloom-level distribution across all 980 items.

B Question-Generation Prompt Pipeline

Figure 7 shows the hierarchical prompt stack used to compose each benchmark question from the taxonomy leaf, Bloom level, and verse context.

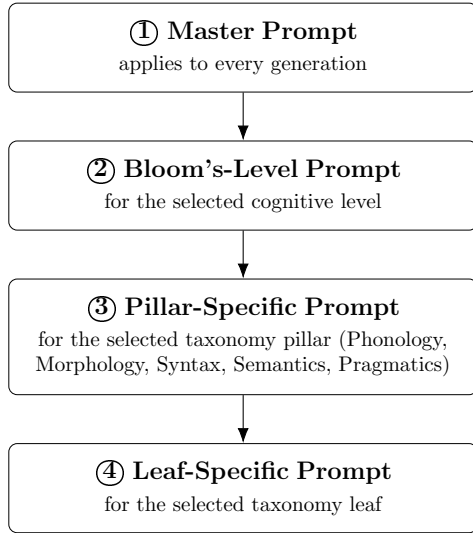


Figure 7: Hierarchical prompt-composition pipeline used to generate each benchmark question.

C Question-Quality Rubric

Every generated question is rated 1–5 on four criteria to guide human review (Figure 8).

D Human Review Interface

Figure 9 shows the custom web interface used for human review and adjudication. For each item, reviewers see the question with the two LLM judges' answers compared against the generator's reference (Figure 9a), the grounding evidence drawn from the source datasets (Figure 9b), and an adjudication panel for recording the final decision (Figure 9c).

E LLM-Judge Scoring Prompt

Open-ended answers are scored automatically by an LLM judge against the reference answer, using the prompt in Figure 10.

F Verifier Guidelines for Bloom Levels

Reviewers assign each Bloom level by the cognitive operation the question demands of the solver, not by how advanced its topic sounds.

Verifier guidelines for assigning Bloom levels

Remember: Recall an explicit fact from the verse or a standard reference, such as a root or a gloss.

Understand: Explain or paraphrase a meaning or grammatical function in one's own words.

Apply: Use a known rule or procedure on a specific instance, such as applying a tajwīd or grammar rule to a given word.

Analyze: Break the verse down and derive a relationship or structure that is not stated, over more than one step.

Evaluate: Weigh competing valid readings and justify which is stronger.

The Bloom level was the most frequently corrected field during human review (235 items; Section 3.3), so the generated labels were verified and corrected rather than accepted as produced.

G Representative Questions by Pillar

One representative question per pillar, with its human-verified Bloom level. Arabic is transliterated; each expected answer follows the arrow.

One question per pillar with its verified Bloom level

Morphology (Remember): What is the triliteral root of the verb *ja'ala*? → j-'-l.

Semantics (Understand): Explain the difference in meaning between *al-raḥmān* and *al-raḥīm* in the *basmala*. → Both derive from the root r-ḥ-m (mercy); *al-raḥmān* is the intensive form denoting all-encompassing mercy to every creature, while *al-raḥīm* denotes sustained, particular mercy.

Phonology (Apply): Apply the *nūn sākinah* rules to *huda min rabbihim* (Q 2:5): what ruling governs the *nūn* of *min* before *rabbihim*, and why? → *Idghām* (assimilation) into the *rā'* with no *ghunnah*, since *rā'* is an *idghām* letter that carries no nasalization.

Syntax (Analyze): In *iiyāka na'budu* (Q 1:5), identify the role of *iiyāka* and explain what its position contributes. → It is a fronted direct object (*maf'ūl bihi muqaddam*); placing the object before the verb restricts worship to God alone.

Pragmatics (Evaluate): In Q 3:7, assess the two accepted stops, pausing at *illā Allāh* versus continuing to *wa-l-rāsikhūna fī l-'ilm*, and what each implies. → The first reading restricts knowledge of the *mutashābih* to God, the second extends it to those firmly grounded in knowledge; both are grammatically valid and attested.

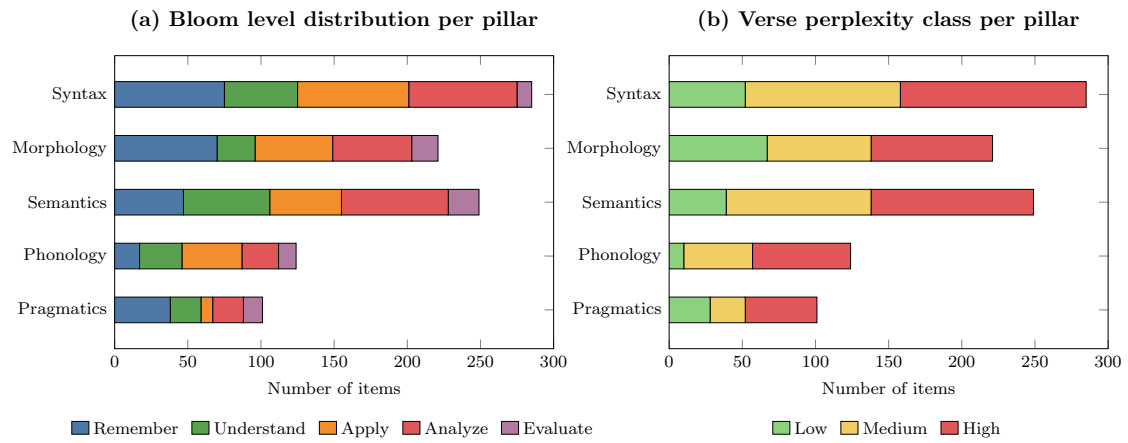


Figure 4: Dataset composition: (a) Bloom-level distribution per pillar and (b) verse-perplexity class per pillar. The overall Bloom-level distribution across all 980 items is given in Figure 3.

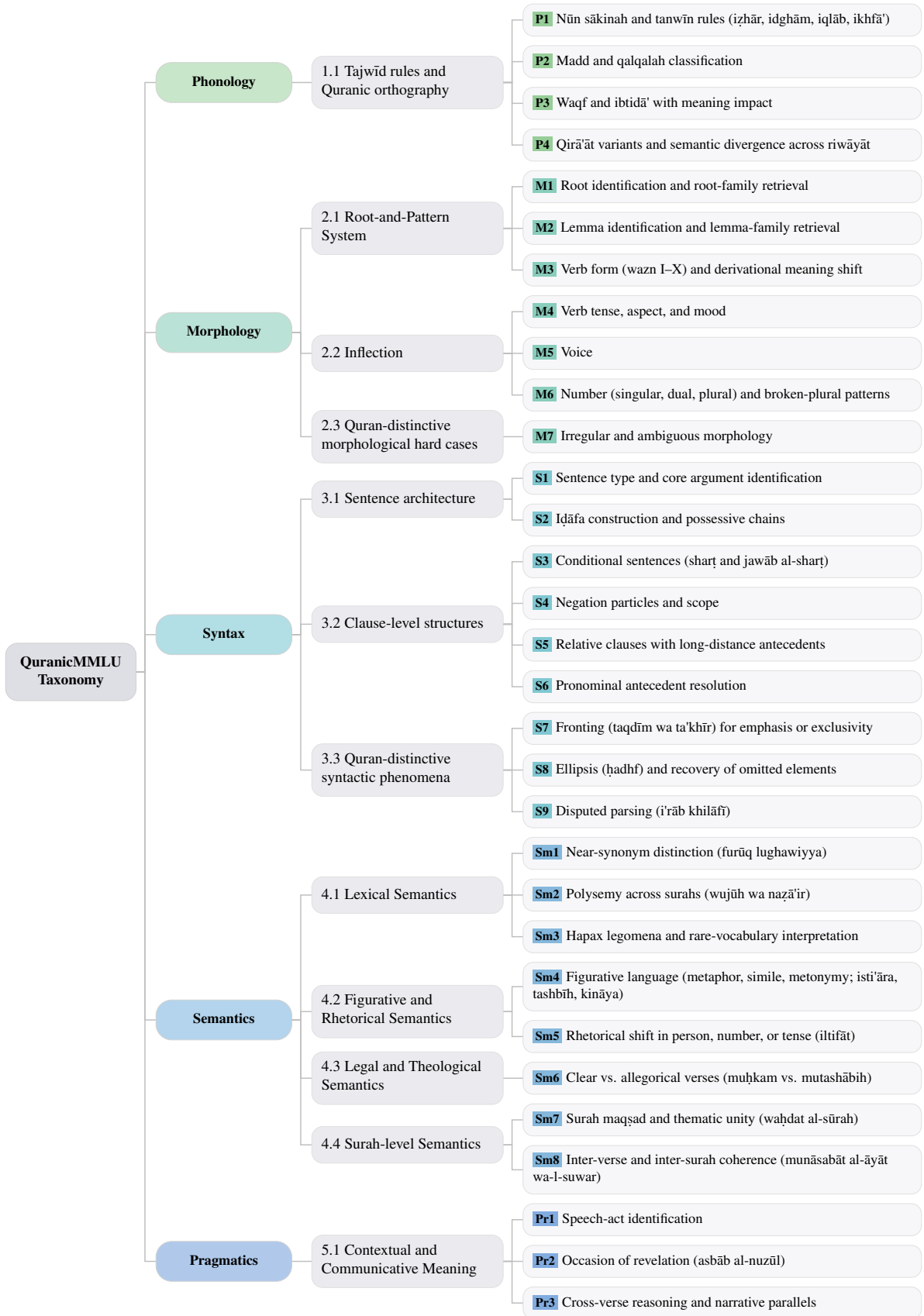


Figure 5: The QuranicMMLU taxonomy: 5 linguistic pillars, 12 categories, and 31 leaves. Leaf codes (e.g., P1, Sm8) index the specific phenomena evaluated by the benchmark.

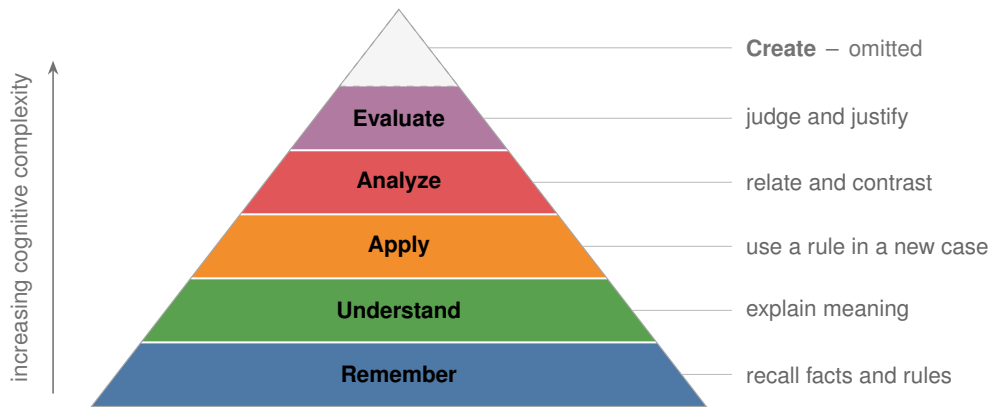


Figure 6: We stratify benchmark questions across five cognitive levels from Bloom’s revised taxonomy (Remember through Evaluate), excluding the Create level, as synthesizing novel Quranic content would violate theological and scholarly principles in Islamic tradition.

Question-quality rubric used during annotation and review

Specificity. Does the question point to exactly one defensible answer? 5 = fully constrained (target word, recitation, and referent fixed or inferable); 1 = many defensible answers, none privileged by the question.

Answerability. Can the question be answered from the question, verse, translation, and leaf alone? 5 = fully self-contained; 1 = depends on an artefact the model never sees, which marks the question as unanswerable from the provided context.

Leaf relevance. Does the question genuinely test its assigned leaf? 5 = squarely on the leaf; 1 = belongs to a different leaf.

Clarity. Is the wording unambiguous, so that two competent readers interpret it identically? 5 = a single interpretation; 1 = incoherent wording.

Figure 8: Question-quality rubric used to guide human review.

Review
Dashboard
Taxonomy guide
Reviewer guide

Morphology
M1 — Root identification and root-family retrieval
Guide
Verse 36:83
Original Bloom remember
q_id79y36d

What is the trilateral root of the noun malakūt in the verse **فَسُبْحَانَ الَّذِي يَدُوهٖ مَلَكُوتُ كُلِّ شَيْءٍ وَإِلَيْهِ تُرْجَعُونَ** (Q 36:83)?

Reviewed by: Admin

فَسُبْحَانَ الَّذِي يَدُوهٖ مَلَكُوتُ كُلِّ شَيْءٍ وَإِلَيْهِ تُرْجَعُونَ

So exalted is He in whose hand is the realm of all things, and to Him you will be returned

GROUND-TRUTH VERSE EVIDENCE — THE DATASET THE ANNOTATORS WERE SHOWN

TASK 1 · SHORT ANSWER — VS ORIGINAL ANSWER & EACH OTHER

ORIGINAL ANSWER
م-ل-ك

GPT (GPT-5.2)
= ORIGINAL
م-ل-ك

Quranic Arabic Corpus token Word 4: LEM:malakuwt ROOT:mlk

GEMINI (2.5-FLASH)
≠ ORIGINAL
m-l-k

Quranic Arabic Corpus (Dukes, 2009/2011)

TASK 2 · MCQ — ORIGINAL CORRECT = B

A. ا-ل-م

B. م-ل-ك **original**

C. ر-ج-ع

D. ل-ك-و

Annotator	Picked	vs Original	Rewritten / note
GPT (gpt-5.2)	B	✓	Quranic Arabic Corpus (36:83), Word 4 مَلَكُوتُ: ROOT mlk.
Gemini (2.5-flash)	B	✓	The Quranic Arabic Corpus analysis for 'malakuwtu' explicitly states 'ROOT:mlk'.

TASK 3 · BLOOM LEVEL — ORIGINAL = REMEMBER

Annotator	Bloom	vs Original	Justification
GPT (gpt-5.2)	remember	✓	The question requires recalling the canonical trilateral root of a given word (malakūt) without further explanation or rule-based analysis.
Gemini (2.5-flash)	Remember	✓	The question asks for the trilateral root of a noun, which is a direct retrieval of a canonical morphological fact without requiring inference, comparison, or conscious application of a multi-step rule.

TASK 4 · QUESTION QUALITY RATINGS

Criterion	GPT (gpt-5.2)	Gemini (2.5-flash)	Δ
specificity	5	5	0
answerability	5	5	0
leaf relevance	5	5	0
clarity	5	5	0
overall score	5	5	0.00

(a) Annotation view: the question with its four tasks (short answer, MCQ, Bloom level, and quality ratings), comparing each LLM judge against the reference answer.

Figure 9: The custom web interface used for human review and adjudication.

GROUND-TRUTH VERSE EVIDENCE — THE DATASET THE ANNOTATORS WERE SHOWN

Verse 36:83

MORPHOLOGY (WORD-BY-WORD) Quranic Arabic Corpus

WORD 1 فَسُبْحٰنَ

فَ prefix · fa- resumption (fa- / wa-, "so / and then")

سُبْحٰنَ stem · noun · masculine · accusative · root س · ب · ح · lemma سُبحن

WORD 2 اَلَّذِیْ

اَلَّذِیْ stem · relative pronoun · masculine singular · lemma اَلَّذِیْ

WORD 3 بِیَدِیْهِ

بِ prefix · preposition bi- ("with / by")

یَدِیْ stem · noun · feminine singular · genitive · root ی · د · ی · lemma ید

هِ suffix · personal pronoun · 3rd person masculine singular

WORD 4 مَلٰکُوتُ

مَلٰکُوتُ stem · noun · masculine · nominative · root م · ل · ک · lemma مَلٰکُوت

WORD 5 کُلِّیْ

کُلِّیْ stem · noun · masculine · genitive · root ک · ل · ل · lemma کُلّ

WORD 6 شَیْءٌ

شَیْءٌ stem · noun · masculine · indefinite · genitive · root ش · ی · ا · lemma شَیْء

WORD 7 وَاِلَیْهِ

وَ prefix · wa- coordinating ("and")

اِلَیْ stem · preposition · lemma اِلَیْ

هِ suffix · personal pronoun · 3rd person masculine singular

WORD 8 تُرْجَعُوْنَ

تُرْجَعُوْنَ stem · verb · imperfective (present) · passive voice · 2nd person masculine plural · root ر · ج · ع · lemma رجع

وْنَ suffix · personal pronoun · 2nd person masculine plural

WORD GLOSSES

fasub'hāna So glory be فَسُبْحٰنَ

alladhī (to) the One who اَلَّذِیْ

biyadihi in Whose hand بِیَدِیْهِ

malakūtu is (the) dominion مَلٰکُوتُ

kulli (of) all کُلِّیْ

shayin things شَیْءٌ

wa-ilayhi and to Him وَاِلَیْهِ

tur'ja'ūna you will be returned تُرْجَعُوْنَ

TRANSLATIONS (6 MOST-USED ACADEMIC)

Sahih International So exalted is He in whose hand is the realm of all things, and to Him you will be returned.

Arberry So glory be to Him, in whose hand is the dominion of everything, and unto whom you shall be returned.

Yusuf Ali So glory to Him in Whose hands is the dominion of all things; and to Him will ye be all brought back.

(b) Grounding view: the word-by-word morphological evidence and translations shown to reviewers as ground truth for each verse.

Figure 9: (continued) Human review and adjudication interface.

ADJUDICATION
q_id79y36d

Verdict:
Keep
Revise
Discard

Best answer:
Original
GPT
Gemini
Custom

Question text — edit to correct wording (blank = keep original)

What is the trilateral root of the noun malakūt in the verse فَسُبْحَنَ الَّذِي يَبْدُءُ الْمَلَكُوتَ كُلَّ شَيْءٍ وَإِلَيْهِ تُرْجَعُونَ (Q 36:83)?

Final / corrected short answer (Task 1)

leave blank to use the chosen source above

MCQ options — edit any text, mark the correct one, or add a new option

☐ A م-ل-ا
☐ B م-ل-ك
☐ C ر-ج-ع
☐ D ل-ك-و

+ Add option

Final Bloom level

remember (original)

Notes

adjudication notes...

Save & mark reviewed
Flag for discussion
Clear
Save → Next

(c) Adjudication panel: reviewers record a verdict (keep, revise, or discard), select or edit the final answer and options, set the Bloom level, and leave notes.

Figure 9: (continued) Human review and adjudication interface.

LLM-judge scoring prompt for open-ended answers

Task. Score each model’s free-form answer against the reference answer for an Arabic Quranic question.

Scoring rubric (0–5), applied strictly. **5 – Fully correct:** same meaning as the reference, no meaningful omission; paraphrase and synonyms are fine. **4 – Mostly correct:** core answer right, one minor detail missing or imprecise; would not mislead. **3 – Partially correct:** at least one correct key fact, but omits or contradicts another. **2 – Mostly wrong:** predominantly incorrect, but contains at least one relevant element. **1 – Barely relevant:** almost entirely wrong, but addresses the right topic or domain. **0 – Wrong / empty:** factually wrong, hallucinated, off-topic, refused, or empty.

Label mapping. correct = 4–5; partial = 2–3; incorrect = 0–1.

Judging notes. Judge meaning, not surface wording; Arabic synonyms and paraphrases are acceptable. For a short entity or term, a near-exact match scores 5 and the correct meaning in other words scores 4. Do not penalise markdown formatting. Correct but irrelevant extra information is not penalised unless it contradicts the reference.

Inputs. the question, the reference answer, and a block of model answers.

Output. a single JSON object (no markdown fences) mapping each model key to a score, label, and a one-sentence reason citing specific content:

```
{
  "model_key": {
    "score": 0-5,
    "label": "correct|partial|incorrect",
    "reason": "..."
  }
}
```

Figure 10: LLM-judge scoring prompt used to grade open-ended answers.